

Channel-Separated Input Encodings for Multi-Signal Transformers: Block Partitioning, Soft Orthogonality, and What the FDM Analogy Buys

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Abstract

Transformers consuming several time-aligned scalar signals (e.g. multivariate sensor streams or multi-modal feature channels) typically embed each channel independently and *sum* the embeddings before self-attention. We show that this default is structurally capped on tasks where channel identity matters: on a controlled synthetic benchmark with non-linear lagged cross-channel interactions, a naive-sum encoder sits at ~ 3.25 validation negative log-likelihood (NLL) regardless of parameter budget ($15\times$ scaling does not help), while two lightweight fixes both reach ~ 2.42 NLL. The first, **concat**, reserves a non-overlapping block of embedding dimensions per channel and concatenates them. The second, **sum-ortho**, keeps summation but adds a soft orthogonality regulariser $\lambda \sum_{i \neq j} (W_i \cdot W_j)^2$ on the per-channel learned projections. Both cut the NLL gap to the uniform baseline by $\sim 84\%$ and more than double top-1 bin accuracy over naive sum; they match each other to within seed std. We also evaluate a frequency-division-multiplexing-inspired variant (**fdm**), which amplitude-modulates a channel-specific sinusoidal carrier inside each block: FDM converges faster but saturates at a worse ceiling (NLL 2.85), with a grid search over carrier bands confirming the gap is structural, not a hyperparameter choice. Channel-as-token and channel-independent architectural baselines both underperform at matched parameter budget. A linear-probe diagnostic shows that block partitioning and soft orthogonality both yield input embeddings from which each raw channel is recoverable to $R^2 \approx 1.0$, preserved through attention to $R^2 \approx 0.93$ at the final layer; naive summation collapses to $R^2 \approx 0.24$ at every layer. We conclude that channel identity is what matters, and two cheap mechanisms—hard partitioning or soft orthogonality—are both adequate ways to supply it. Code: <https://github.com/OssiLehtinen/FDM-encoding>.

1 Introduction

A canonical use of transformers outside language is fitting a sequence model to several time-aligned numerical signals: multi-sensor telemetry, financial factor stacks, biomedical waveforms, multi-modal embeddings along time. The standard recipe inherited from vision and language (1) maps each scalar input to a vector and *adds* those vectors together—the same trick used to inject positional information. We will refer to this as the *sum* encoding.

Sum encoding has a structural problem when the input is multiple *simultaneous* channels rather than a single token-plus-position pair. Given $\mathbf{h} = \sum_k W_k x_k$, the network cannot recover individual x_k from $\mathbf{h}$ without learning to implicitly allocate orthogonal subspaces—a constraint that nothing

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in the loss function explicitly enforces. Our empirical finding is blunt: on a task where channel identity is informative, the optimiser does not, in practice, allocate those subspaces on its own, even with $15\times$ more parameters than are needed by alternatives.

We study two cheap mechanisms that supply channel identity and one inspired-by-communications mechanism that does not quite work as advertised:

- **Block partitioning** (CONCAT): reserve a d_{model}/C -dim block per channel and concatenate. Channel identity is a hard structural property.
- **Soft orthogonality** (SUM-ORTHO): keep naive summation, but add $\lambda \sum_{i \neq j} (W_i \cdot W_j)^2$ to the loss so the per-channel projections are pulled toward orthogonal directions. Channel identity is a learned structural property.
- **Frequency-division-multiplexing** (FDM): within each partition block, amplitude-modulate a sinusoidal carrier at a channel-specific frequency, in the spirit of rotary position embedding (RoPE) (2).

We compare them against SUM and against two architectural baselines borrowed from the time-series forecasting literature: CI (channel-independent, PatchTST-spirit (5)) and CAT (channel-as-token, iTransformer-spirit (6)).

Key findings. (i) CONCAT and SUM-ORTHO are essentially tied and dominate everything else on the benchmark; (ii) FDM’s frequency carrier is a *faster-convergence bias* but a *worse ceiling* than block partitioning alone, because the fixed template has fewer degrees of freedom than a learned projection once the block is wide enough to exploit; (iii) the architectural baselines CI and CAT do not close the gap at matched parameter budget; (iv) a linear-probe diagnostic, a test-time channel-mask ablation, and a model-width scaling study consistently rank encoders the same way as the downstream metric.

Contributions.

- A controlled benchmark with lagged cross-channel interactions designed to expose whether an encoder preserves channel identity.
- A six-way comparison (SUM, SUM-ORTHO, CONCAT, FDM, CI, CAT) at matched sequence length and total parameter budget, over 5 seeds with 100-epoch cosine-scheduled training.
- Evidence that SUM-ORTHO is the cheapest retrofit for legacy sum-encoding pipelines: no architectural change, ~ 200 extra parameters, same ceiling as hard block partitioning.
- A reframing of FDM-style carrier encodings as a convergence-speed inductive bias rather than a better ceiling, supported by a width sweep ($d_{\text{model}} \in \{64, 128, 256\}$) and an 11-point carrier-band grid search.

2 Method

2.1 Problem setup

We are given a sequence $\{x_t\}_{t=1}^T$ where $x_t \in \mathbb{R}^C$ is a vector of C scalar channels at time t . We want a transformer to model $p(y_t \mid x_{\leq t-1})$ where y_t is a downstream target, binned into K classes for a categorical generative head. The question we address is purely about how x_t enters the residual stream.

2.2 Encoders

sum. Baseline:

$$\mathbf{h}(t) = \sum_k (W v_k(t) + e_k) + \mathbf{p}(t),$$

with shared scalar projection $W \in \mathbb{R}^{d_{\text{model}} \times 1}$, learned channel embedding e_k , and sinusoidal positions $\mathbf{p}(t)$.

sum-ortho. Like SUM but with a per-channel learned projection $W_k \in \mathbb{R}^{d_{\text{model}}}$ and an auxiliary orthogonality loss

$$L_{\text{ortho}} = \lambda \sum_{i \neq j} (W_i \cdot W_j)^2 / 2,$$

added to the task NLL; we use $\lambda=10^{-2}$. Channel-separated subspaces are a soft constraint, overridable by the task loss if necessary.

concat. Reserve $d_{\text{block}}=d_{\text{model}}/C$ dims per channel and concatenate:

$$\mathbf{e}_k(t) = W_k v_k(t) + b_k, \quad \mathbf{h}(t) = [\mathbf{e}_0(t) \parallel \cdots \parallel \mathbf{e}_{C-1}(t)] + \mathbf{p}(t).$$

Channel identity is a hard structural property at the input layer.

fdm. Like CONCAT, but the per-block projection is replaced by a sinusoidal carrier amplitude-modulated by the scalar value. For channel k at time t , with base $b=10000$ and carrier ω_k log-spaced across $[0.5, 8.0]$:

$$\mathbf{e}_k(t)_{2i} = v_k(t) \sin(\omega_k t b^{-2i/d_{\text{block}}}), \quad \mathbf{e}_k(t)_{2i+1} = v_k(t) \cos(\omega_k t b^{-2i/d_{\text{block}}}), \quad (1)$$

for $i = 0, \dots, d_{\text{block}}/2 - 1$. The encoder is parameter-free (learnable ω_k is evaluated as a separate variant, FDM-LEARN). Attention dot products between two positions in the same channel block become $v_k(t_1)v_k(t_2) \langle \phi(t_1), \phi(t_2) \rangle$, exposing lag-dependent intra-channel correlations directly to the softmax—a RoPE-style intra-channel positional prior.

ci (channel-independent). PatchTST-spirit (5): one causal transformer is run with shared weights over each channel’s univariate sequence; final hidden states at each time are concatenated into the head. Each channel gets full d_{model} capacity; compute scales as $\mathcal{O}(CBT^2)$.

cat (channel-as-token). iTransformer-spirit (6): each (t, k) pair becomes its own token, sequence length is $C \cdot T$, and a standard transformer runs on the flattened stream with a causal-by-time mask (tokens at $(t, \cdot)$ cannot see tokens at $(t', \cdot)$ for $t' > t$). Prediction at time t reads from the last-channel token of time t . Attention cost is $\mathcal{O}((CT)^2)$; we skip this baseline at $C=16$.

2.3 Dataset

We synthesise $N=512$ time series of length $T=160$ with C channels each. Each channel is an independent stochastic process: a sum of three sinusoids with random frequencies in $[0.005, 0.08]$ cycles/sample (angular $\approx [0.03, 0.50]$ rad/sample), random phases, and first-order autoregressive

Table 1: Main predictive comparison at $C=4$, $d_{\text{model}}=64$. 100 epochs with cosine decay, mean $\pm$ std over 5 seeds; metric is the best val NLL achieved during training (effective early stopping). Random baseline: $\text{NLL}=\ln 32 \approx 3.466$, $\text{acc}=0.031$. All models share the same 3-layer backbone at $\sim 152\text{K}$ params; encoder rows are ordered by best val NLL (lower is better).

Encoder	Class	Enc. params	Val NLL $\downarrow$	Val acc $\uparrow$
SUM (baseline)	summation	320	3.253 ± 0.007	0.092 ± 0.003
CAT (channel-as-token)	arch.	320	3.088 ± 0.040	0.122 ± 0.008
CI (channel-independent)	arch.	2,112	3.051 ± 0.008	0.140 ± 0.003
FDM (block + carrier)	block	0	2.854 ± 0.018	0.145 ± 0.004
concat (block, no carrier)	block	128	2.429 ± 0.038	0.201 ± 0.007
sum-ortho ($\lambda=10^{-2}$)	sum + reg	512	2.422 ± 0.021	0.202 ± 0.004

[AR(1)] Gaussian noise; signals are then standardised. The outcome is a fixed non-linear function of the *first four* channels with two different lags,

$$y_t = \tanh(s_0(t-3) s_1(t)) + 0.6 \sin(1.3 s_2(t-7)) + 0.4 \mathbb{I}[s_3(t) > 0] s_0(t),$$

binned into $K=32$ quantile bins. Any additional channels ($k \geq 4$) are independent distractors. The signal content occupies $[0.03, 0.5]$ rad/sample; FDM carriers live in $[0.5, 8.0]$ rad/sample, so the FDM gain is not a trivial resonance with the data.

2.4 Architecture and training

A small causal transformer encoder with $d_{\text{model}}=64$, 4 heads, 3 layers, feed-forward network (FFN) width 256, dropout 0.1, pre-LayerNorm. The head is a linear map to $K=32$ logits. Training: AdamW, lr 3×10^{-4} with cosine decay to 1% of peak, weight decay 10^{-4} , batch size 32, gradient clip 1.0, 100 epochs. Each result is the mean over 5 random seeds (3 for the carrier grid). Metrics are reported at the best val NLL achieved during training (effective early stopping); trajectories are saved so convergence can be audited.

3 Results

3.1 Main comparison

Table 1 reports all six encoders at $C=4$, $d_{\text{model}}=64$. The ordering is consistent across all five seeds. Two cheap fixes—CONCAT (hard block partitioning) and SUM-ORTHO (soft orthogonality on learned projections)—sit on top, tied to within one seed std (~ 0.02 NLL). Both more than double SUM’s bin accuracy. FDM sits ~ 0.4 nats worse than either, still clearly beating SUM. The two architectural baselines CI and CAT underperform all of them at the same parameter budget, despite giving each channel full d_{model} capacity (ci) or $C \cdot T$ sequence length (cat).

3.2 Scaling with channel count

All encoders degrade as C grows, because channels $k \geq 4$ are distractors that add noise to the prediction target. The top two remain SUM-ORTHO and CONCAT at every C . The SUM-ORTHO—CONCAT tie is stable: the soft regulariser continues to achieve what hard partitioning does, which is notable because SUM-ORTHO does not give the optimiser a fixed allocation—it must discover orthogonal W_k by itself.

Table 2: Scaling with channel count. Channels 0–3 drive the outcome; additional channels are independent distractors. 100 epochs with cosine decay, mean $\pm$ std over 5 seeds. CAT is skipped at $C=16$ because its $\mathcal{O}((CT)^2)$ attention cost explodes.

C	encoder	val NLL $\downarrow$	val acc $\uparrow$
4	SUM	3.253 ± 0.007	0.092 ± 0.003
	CAT	3.088 ± 0.040	0.122 ± 0.008
	CI	3.051 ± 0.008	0.140 ± 0.003
	FDM	2.854 ± 0.018	0.145 ± 0.004
	concat	2.429 ± 0.038	0.201 ± 0.007
	sum-ortho	2.422 ± 0.021	0.202 ± 0.004
8	SUM	3.302 ± 0.005	0.080 ± 0.002
	CAT	3.103 ± 0.083	0.114 ± 0.010
	CI	3.054 ± 0.014	0.135 ± 0.004
	FDM	2.972 ± 0.027	0.135 ± 0.005
	CONCAT	2.515 ± 0.020	0.187 ± 0.008
	sum-ortho	2.437 ± 0.029	0.200 ± 0.009
16	SUM	3.309 ± 0.004	0.075 ± 0.003
	CI	3.074 ± 0.023	0.132 ± 0.006
	FDM	3.030 ± 0.011	0.126 ± 0.006
	CONCAT	2.651 ± 0.028	0.169 ± 0.006
	sum-ortho	2.566 ± 0.044	0.182 ± 0.009

3.3 Scaling with model width

Table 3 and Figure 2 vary d_{model} over $\{64, 128, 256\}$ with $d_{\text{ff}}=4d_{\text{model}}$. The picture is very clean. SUM is structurally capped: total parameter count grows $15\times$ across the sweep, and validation NLL changes by less than 0.02. No amount of width fixes rank-deficient encoding. CONCAT scales strongly from 2.429 at $d=64$ down to 2.291 at $d=128$ before mild overfitting at $d=256$. FDM plateaus between 2.85 and 2.93 and overfits hard at $d=256$ (best epoch 10 of 100). The gap between CONCAT and FDM widens with width, consistent with the fixed sinusoidal template becoming a ceiling constraint once the block has room to learn a richer projection.

Is it the frequency choice? We ran two controls at $d_{\text{model}}=128$. (i) A FDM-LEARN variant with ω_k as learnable parameters (initialised at the log-spaced defaults) changes NLL and accuracy by less than 10^{-3} : diagnostics show the ω_k drift by $< 1\%$ under AdamW in 100 epochs because the gradient signal to four scalar frequencies is dominated by the surrounding Q/K/V/FFN weights. (ii) A grid search over 11 fixed carrier bands ($\omega_{\min}, \omega_{\max}$) spanning $[0.03, 0.5]$ (matching the signal bandwidth) through $[10, 100]$ yields a best NLL of 2.809 ± 0.003 for the signal-matched band—better than the default 2.900 by ~ 0.09 nats, but still 0.52 nats behind CONCAT’s 2.291. *No fixed carrier band closes the gap.*

3.4 Channel identifiability via linear probing

We freeze a trained model and fit closed-form ridge probes $\mathbf{h} \mapsto \hat{x}$ from hidden state at layer ℓ to the raw input x , reporting held-out per-channel R^2 (Table 4). Three encoders achieve near-perfect

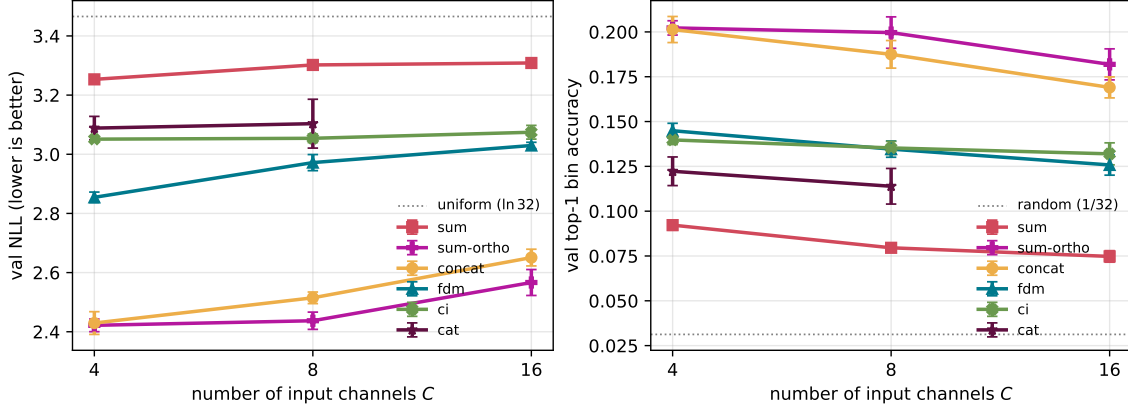


Figure 1: Scaling with the number of input channels C . Error bars are one seed std over 5 seeds. All encoders degrade with C (more distractor channels), but CONCAT and SUM-ORTHO maintain a wide margin over SUM. CAT is skipped at $C=16$ due to $\mathcal{O}((CT)^2)$ attention cost.

recovery at the input layer: FDM by construction (Eq. 1), CONCAT because each W_k converges to a non-degenerate map, and SUM-ORTHO because the orthogonality regulariser drives the learned W_k toward mutually orthogonal directions. At layer 3, all three still admit mean $R^2 \geq 0.92$; SUM collapses to $R^2 \approx 0.24$ at every depth. This is a clean recovery-versus-ceiling correspondence: the encoders whose residual stream preserves channel identity through attention are the same encoders that win on the downstream task.

3.5 Robustness to missing channels

At test time we zero one channel at a time and re-evaluate (Table 5). CONCAT and SUM-ORTHO exhibit the sharpest and most interpretable per-channel signatures: masking ch0 (which appears in two interaction terms in the outcome) drops accuracy by ~ 0.11 , masking ch1 by ~ 0.09 , ch2 by ~ 0.08 , and ch3 (lightest involvement) by ~ 0.01 . FDM is noticeably less sharp (Δ_{ch0} only -0.062). The architectural baselines CI and CAT give partial per-channel structure but with less clean ordering; SUM is nearly flat.

4 Discussion

Block partitioning vs. soft orthogonality. CONCAT and SUM-ORTHO are tied on every metric we measured—downstream NLL, accuracy, C -scaling, probe recoverability, mask localisation. They are quite different mechanisms: one hard-codes a non-overlapping subspace allocation at the architectural level; the other lets the optimiser discover orthogonal projections under a gentle penalty. The fact that both reach the same operating point suggests that *channel identity is what matters, and several mechanisms for supplying it are equivalent*. SUM-ORTHO is likely the more practically useful of the two: it is a drop-in addition to existing sum-encoding pipelines (one regulariser term, $\sim C \cdot d_{\text{model}}$ extra parameters), with no architectural change and no hard dimension-budget constraint tying d_{model} to a divisor of C .

What the FDM carrier buys (and doesn’t). FDM is not what it sounds like. In communications, frequency-division multiplexing prevents cross-talk on a *shared* medium; in our setting the

Table 3: Scaling with d_{model} at $C=4$. $d_{\text{ff}}=4d_{\text{model}}$, heads=4 (so d_{head} grows). Mean $\pm$ std over 5 seeds, 100 epochs with cosine decay. SUM shows its structural ceiling: $15\times$ more parameters barely move it. CONCAT scales strongly but begins to overfit at $d_{\text{model}}=256$. FDM plateaus and overfits earlier because amplitude-modulating a fixed template is a constrained representation that large blocks cannot fully exploit.

d_{model}	encoder	total params	val NLL $\downarrow$	val acc $\uparrow$
64	SUM	152,480	3.253 ± 0.007	0.092 ± 0.003
	FDM	152,160	2.854 ± 0.018	0.145 ± 0.004
	FDM-LEARN	152,164	2.856 ± 0.018	0.145 ± 0.005
	concat	152,288	2.429 ± 0.038	0.201 ± 0.007
128	SUM	599,840	3.251 ± 0.008	0.094 ± 0.002
	FDM	599,200	2.898 ± 0.007	0.146 ± 0.003
	FDM-LEARN	599,204	2.899 ± 0.007	0.145 ± 0.003
	concat	599,456	2.291 ± 0.009	0.223 ± 0.002
256	SUM	2,379,296	3.255 ± 0.007	0.091 ± 0.001
	FDM	2,378,016	2.930 ± 0.015	0.139 ± 0.003
	FDM-LEARN	2,378,020	2.930 ± 0.015	0.140 ± 0.003
	concat	2,378,528	2.322 ± 0.014	0.221 ± 0.003

blocks do not overlap, so there is no cross-talk to prevent in the first place. What the sinusoidal carrier actually provides is an intra-block RoPE-style positional prior: attention dot products between two positions in the same channel gain a lag-dependent cos factor without needing the model to learn the demix from the additive position channel. Our experiments show this is a *convergence-speed advantage*: FDM reaches a given NLL sooner than CONCAT early in training (faster descent), but is eventually overtaken. At convergence CONCAT wins cleanly, and at larger d_{model} the gap widens because the fixed template cannot use the extra width the way a learned projection can. A grid search over 11 fixed carrier bands and a learnable- ω_k variant both fail to close this gap; the constraint is structural, not a hyperparameter miss.

Architectural baselines. CI and CAT give each channel nominally more capacity than the block-partitioned encoders (full d_{model} per channel and per- (t, k) token respectively) but both underperform at matched parameter count. CI shares a single backbone across channels; the bottleneck is the final-layer head’s limited capacity to combine C per-channel streams. CAT pays $\mathcal{O}((CT)^2)$ attention and gains nothing for it on this task, presumably because the relevant structure is simultaneity of channels at a given t , not long-range temporal dependencies across channels at different times.

Why sum is structurally capped. Our SUM baseline, under $15\times$ parameter scaling, moves validation NLL by less than 0.02 nats. We interpret this not as a training failure but as the absence of any gradient signal that would encourage the optimiser to allocate orthogonal subspaces on its own. A soft orthogonality penalty, which makes that allocation explicit, immediately recovers the full performance gap.

LayerNorm does not destroy channel separation. Standard pre-LayerNorm could in principle mix channel blocks (it computes statistics across the full d_{model} axis), but empirically the linear

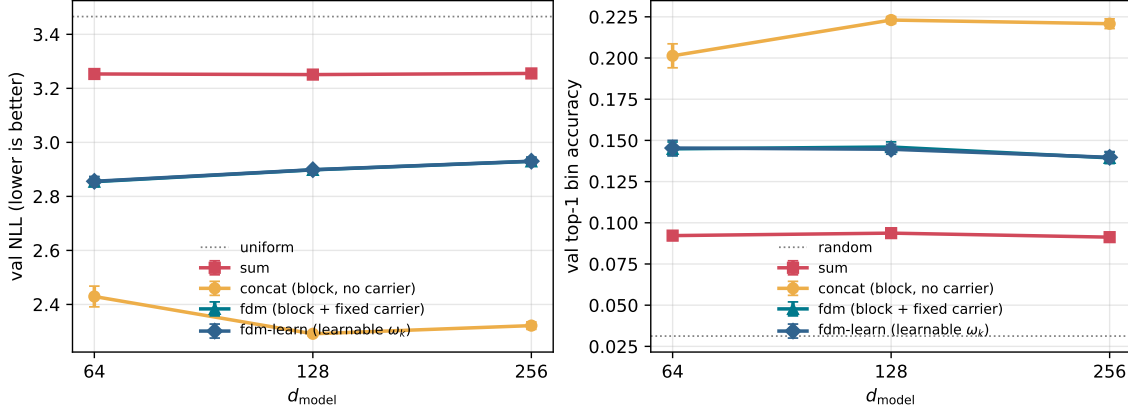


Figure 2: Scaling with d_{model} at $C=4$. SUM is essentially flat: extra width buys nothing because the bottleneck is structural, not representational. CONCAT improves and eventually overfits at $d_{\text{model}}=256$. FDM plateaus earlier because amplitude-modulating a fixed template has fewer degrees of freedom than a learned projection once $d_{\text{block}} \geq 32$.

probes at layer 3 recover each channel to $R^2 \geq 0.92$. LayerNorm rescales each coordinate by a statistic derived from all coordinates but does not linearly combine them; the directional block structure survives.

5 Limitations

- The benchmark is synthetic and deliberately constructed to make channel identity informative. On tasks where channels are truly exchangeable or redundant, SUM may not be penalised.
- Scales tested: $d_{\text{model}} \in \{64, 128, 256\}$, $C \in \{4, 8, 16\}$, up to $\sim 2.4\text{M}$ parameters. At larger model scales the interplay between encoders and overfitting may change.
- The CAT baseline is skipped at $C=16$ for compute reasons; comparing it at matched attention budget rather than matched parameters would be a fairer test.
- Carrier hyperparameters for FDM were searched but we did not tune every design knob (carrier ladder base, per-block LayerNorm). A per-block group-norm variant may alter the FDM ceiling.
- Five seeds per configuration; for the marginal differences inside the top tier, a larger seed budget would tighten confidence intervals.

6 Related work

Channel-as-token and channel-independent TSF. iTransformer (6) transposes the problem so each variate is a token with full d_{model} capacity; PatchTST (5) runs an independent transformer per channel with shared weights; Crossformer (7) does both directions. These approaches all preserve per-channel capacity at the cost of longer sequences or per-channel compute. The block-partitioned encoding studied here sits in a different corner of the design space: C channels share one d_{model} -wide token at each time, with partitioning and identity preserved *within* the embedding rather than across tokens.

Table 4: Linear-probe channel recovery at $C=4$, seed 0, after 100 epochs of cosine-scheduled training. Closed-form ridge probe $\mathbf{h} \mapsto \hat{x}$ from a frozen hidden state to the raw input vector $x \in \mathbb{R}^4$, held-out per-channel R^2 . Layer 0 is the input embedding; layer 3 is the final transformer block. SUM-ORTHO matches block-partitioned encoders, confirming that the orthogonality regulariser drives learned per-channel projections to near-orthogonal subspaces.

Encoder	Layer	R^2_{ch0}	R^2_{ch1}	R^2_{ch2}	R^2_{ch3}	mean
SUM	0 (input)	0.255	0.250	0.242	0.214	0.240
SUM	3 (final)	0.251	0.243	0.235	0.206	0.234
SUM-ORTHO	0 (input)	1.000	1.000	1.000	1.000	1.000
SUM-ORTHO	3 (final)	0.963	0.933	0.934	0.875	0.926
CONCAT	0 (input)	0.995	1.000	1.000	1.000	0.999
CONCAT	3 (final)	0.948	0.950	0.943	0.906	0.937
FDM	0 (input)	1.000	1.000	1.000	1.000	1.000
FDM	3 (final)	0.927	0.941	0.951	0.882	0.925

Rotary position embedding (RoPE). Su et al. (2) encodes positional information with rotation by frequency-graded angles. FDM re-uses the per-dimension frequency ladder but applies it within each channel’s block, with the *channel* carrier setting the outer scale.

Perceiver IO and Fourier-feature input encodings. Jaegle et al. (3) attaches explicit Fourier features per modality to disambiguate inputs to a shared latent attention bottleneck. FDM differs in that the modality identifier and the signal value are *multiplied* (amplitude modulation) rather than concatenated, and dimensions are physically partitioned rather than augmented.

Orthogonality regularisation. Soft-orthogonality penalties on weight matrices have been used for conditioning and feature decorrelation (8). SUM-ORTHO applies the same idea specifically to per-channel input projections; to our knowledge this is a new use case in the multi-signal embedding literature.

Frequency-domain input encoders. TimesNet (4) processes time-series in the frequency domain with an explicit fast Fourier transform (FFT). Our work is dual: we perform an inverse-FFT-like *encoding* of a single value onto a per-channel frequency, rather than transforming the signal itself.

7 Conclusion

On a task where channel identity is informative, summing per-channel embeddings is structurally capped: no realistic amount of width closes the gap. Two cheap mechanisms both supply the missing channel identity and reach the same operating point. Hard block partitioning (CONCAT) is an architectural choice that guarantees it; soft orthogonality on learned projections (SUM-ORTHO) is an additive regulariser that converges to equivalent behaviour. The frequency-division analogy (FDM) provides a faster-convergence-but-lower-ceiling variant: useful as an inductive bias at small capacity, a constraint at larger ones. The architectural alternatives from the time-series forecasting literature (channel-as-token, channel-independent) underperform both at matched parameter budget on this task.

Table 5: Test-time channel ablation at $C=4$, seed 0. We zero a single channel’s input at inference; rows show the resulting val accuracy. Δ is accuracy drop from the unmasked baseline. The outcome formula uses ch0 in two interaction terms, ch1 in one, and ch2/ch3 once each with smaller marginal effect; block-partitioned and orthogonalised-sum encoders reproduce this ordering cleanly, whereas SUM, CI and CAT are much flatter.

encoder	no mask	mask ch0	mask ch1	mask ch2	mask ch3
SUM	0.093	0.078	0.087	0.090	0.093
Δ acc	—	−0.015	−0.006	−0.003	0.000
CAT	0.135	0.088	0.126	0.101	0.128
Δ acc	—	−0.047	−0.009	−0.034	−0.007
CI	0.140	0.088	0.121	0.096	0.141
Δ acc	—	−0.052	−0.019	−0.044	+0.001
FDM	0.148	0.086	0.099	0.118	0.146
Δ acc	—	−0.062	−0.049	−0.030	−0.002
CONCAT	0.200	0.091	0.103	0.119	0.188
Δ acc	—	−0.109	−0.097	−0.081	−0.012
SUM-ORTHO	0.201	0.089	0.109	0.115	0.187
Δ acc	—	−0.112	−0.092	−0.086	−0.014

Of the three channel-separating recipes, SUM-ORTHO is the most practical retrofit: it is a one-line change to existing sum-encoding pipelines, adds $\sim C \cdot d_{\text{model}}$ trainable parameters, preserves architecture compatibility, and matches hard block partitioning on every metric we measured.

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